

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

Xingqiang Chen and Qixing Wang

Department of Physics

Xiamen University

Xiamen 361005, China

(Dated: July 21, 2026)

Substitutional doping and biaxial strain are jointly used to tune covalent molecular networks, yet total energies are still often assembled from separate composition-only and strain-only density-functional theory calculations. We study quasi-hexagonal C₆₀ graphullerene with the PBE+D3 functional and define a synergy order parameter $\mathcal{S}$ (meV/atom) for the total-energy cross term missing from sequential strain–doping scans. On B/N/P tetramers and periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$) under matched PBE+D3, linear strain coefficients α and $|\mathcal{S}|$ decouple on rigid tetramer grids (Table VI), while phosphorus exhibits the largest joint-load cross term on periodic cells, with $|\mathcal{S}|$ reaching 31.9 meV/atom at $n=1$ under +3% biaxial strain. These bounds are *rigid-strain upper bounds* at fixed fractional coordinates, not equilibrium thermodynamic quantities; ionic relaxation can reverse or nearly extinguish $\mathcal{S}$ (tetramer Table V; periodic $n=1$ P), and tetramer α is sensitive to dopant placement at $\sim 5\%$ loading. The results indicate that stability ranking under load in covalent molecular networks requires explicit coupled strain–doping energy evaluations when chemical mismatch or strain magnitude enter the geometric-nonlinear regime, rather than additive composition–strain models used in sequential high-throughput screening.

I. INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many high-throughput pipelines still assemble $E_{\text{tot}}(\epsilon, \delta)$ from *separate* doping-only and strain-only density-functional theory runs [1–3]. Defect formation energies and substitutional stabilities under strain are classical questions in defect physics, but most workflows still assume that composition-only and strain-only increments superpose linearly—an assumption that breaks down when dopant-induced internal stress couples nonlinearly to external load. The omitted cross term is a systematic bias in sequential screening: it arises whenever a dopant introduces a *local chemical-mismatch stress field* that couples nonlinearly to an external biaxial load, so additive composition–strain models can mis-rank stability margins even when dopant ordering at $\epsilon=0$ is unchanged. Synthetic carbon allotropes with extended π conjugation underpin optoelectronic applications; charge transport in molecular networks is often polaronic and anisotropic [4–7], yet prior work emphasizes band edges, mobility, or hopping rather than a periodic total-energy cross term under joint biaxial (ϵ, δ) load. Quasi-hexagonal (qHP) C₆₀ graphullerene [4, 5] provides a covalently linked model network for such coupled tuning, with interfullerene linking that raises hopping couplings relative to vdW C₆₀ monolayers [4]; foundational and recent DFT surveys of qHP monolayers and covalent fullerene networks emphasize elastic, optical, and transport properties [8, 9] but do not report a periodic total-energy cross term under joint biaxial (ϵ, δ) load. Ab initio models of pristine qHP networks report intrinsic intramolecular strain from the eight-bond hexagonal connectivity [10] ($\sim 0.1\text{--}0.2$ eV-scale cage distortions relative to isolated

C₆₀ in prior surveys), so external biaxial load couples to an already pre-strained, *soft* molecular cage—a regime where Eshelby-type linear superposition of defect and external fields is not guaranteed. Beyond pristine monolayers, strain and substitution independently reshape band edges [2, 11, 12], while substitutional B/N engineering can yield spin-polarized half-semiconductor gaps [13], and related 2D fullerene networks show anisotropic mechanical and tribological response under load [14, 15] and endohedral qHP C₆₀ cage filling reshapes band gaps [16]; none of these reports a periodic total-energy cross term under joint biaxial (ϵ, δ) load, motivating explicit coupled grids rather than an additive superposition of composition and strain effects. Non-additive coupling is documented in other two-dimensional networks [1, 17]; a quantitative *total-energy* synergy order parameter $\mathcal{S}$ [Eq. (1)] has not been reported for substitutional B/N/P qHP C₆₀ tetramers and $n = 1\text{--}8$ supercells. Unlike a single-reference formation energy or a pairwise interaction energy, $\mathcal{S}$ is a double-reference cross term over the joint (ϵ, δ) control space that bounds the bias of additive sequential screening (Table I); the novelty is this physically interpreted, validated coupling descriptor rather than the algebraic form of Eq. (1).

We address three questions on this *class* of covalent molecular networks (qHP C₆₀ as the validation platform): (1) how donor vs. acceptor dopants yield distinct mechanical strain response and π filling under the same PBE+D3 protocol; (2) how large the omitted synergy cross term $|\mathcal{S}|$ is under joint load; and (3) whether $|\mathcal{S}|$ scales with size mismatch or instead reflects π -electron nonlinearity. Using $\gtrsim 90$ DFT tasks catalogued in Table III, we define $\mathcal{S}$ on periodic PBE+D3 grids (fifteen (n, δ) points at +3% strain). Matched-functional tetramer α and $\mathcal{S}_{+3\%}^{\text{tet}}$ on an independent PBE+D3 placement map (Table VI) are compared to periodic $\mathcal{S}(n)$ on

the same functional. The largest periodic $|\mathcal{S}|$ at fixed coordinates (up to 31.9 meV/atom for P at $n=1$) does not track tetramer $|\alpha|$ in any simple monotonic sense—*decoupling* here means that linear α cannot quantitatively predict cross-term magnitude or sign, not that $|\alpha|$ and $|\mathcal{S}|$ are anticorrelated. *Local* dopant distortion and *global* biaxial strain interact nonlinearly in total energy: $|\mathcal{S}|$ can exceed what linear α alone would suggest, so *relative* stability margins from $\epsilon = 0$ substitution energies or strain-only elastic increments need not survive a coupled load even if they do not reorder dopants with widely separated E_{sub} values. Protocol limits (ionic-relaxation sign reversal, placement sensitivity) are benchmarked in Sec. IIF; polaron transport diagnostics appear in the Supplemental Material and do not enter $\mathcal{S}$.

II. COMPUTATIONAL METHODS

A. Electronic-structure setup

All periodic audit and charged-state tasks use CP2K [18] with PBE [19] and Grimme DFT-D3 (Becke–Johnson damping) [20]. Table VI supplies verified PBE+D3 tetramer α and $\mathcal{S}_{+3\%}^{\text{tet}}$ on an alternate placement map; the matched PBE+D3 reference map (seed 42; Table III, 24/24) gives +58.7, −297.5, and +456.0 meV/% for B, N, and P (Fig. 1(c)), illustrating placement sensitivity relative to the alternate map. Table III catalogues task groups, N_{conv} , main-text linkage, and protocol; superseded grids are in Table IV. We do not employ the Koopmans-compliant rVV10 approach used in Ref. 4, and instead adopt a PBE+D3 protocol for periodic and charged-state tasks suitable for large-scale parameter scanning, at the cost of absolute accuracy in polaron binding energies.

B. Notation and conventions

Biaxial strain is denoted ϵ (percent); dopant species $\delta \in \{B, N, P\}$. The synergy order parameter is denoted by the calligraphic symbol $\mathcal{S}$ (never the Latin s), with units of meV per atom [Eq. (1)]. π -projected densities of states are written π -DOS (never bare “pi” in equations).

1. Rigid-strain protocol

Biaxial strain scales the in-plane lattice vectors of the *pristine* equilibrium cell at fixed fractional coordinates (no ionic relaxation under load). Reported α and $\mathcal{S}$ under this protocol are upper bounds at fixed coordinates (Sec. IIF).

2. Plane-wave cutoff (size scaling)

Archived CP2K inputs use 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table III); comparisons at fixed n always share the same cutoff. Cross- n trends mix a plane-wave cutoff change at $n = 6$; the cutoff benchmark row in Table III provides a 400 Ry control at $6 \times C_{60}$ (Supplemental Material Table S1 for $n \geq 6$ production rows).

C. Substitutional doping and grids

Tetramer models ($4 \times C_{60}$, 240 sites) substitute B, N, or P at nominal 5% (12–13 sites from archived coordinates); these dopant configurations are not thermodynamically optimized ground states. A second tetramer placement ensemble (24 single-point energies: 18 dopant-strain points plus six pristine references; Table VI) tests whether tetramer α and $\mathcal{S}$ are placement-sensitive at $\sim 5\%$ nominal doping. All periodic supercells propagate the same dopant site map (one substituent per C_{60}); quantitative $|\mathcal{S}|$ on the fifteen-point grid is therefore configuration-specific, whereas the verified PBE+D3 alternate tetramer map (Table VI) gives $|\alpha_P| \gg |\alpha_{B,N}|$ while $\mathcal{S}_{+3\%}^{\text{tet}}$ remains $\mathcal{O}(10)$ meV/atom for all three dopants—so α trends and cross-term magnitudes are placement-specific at $\sim 5\%$ nominal doping (Sec. IIE 2). Periodic scaling cells place one substituent per C_{60} (n dopants among $60n$ lattice atoms, ~ 1.67 at.% for all n), contrasting with the tetramer grid ($\sim 5\%$ nominal). Counting a single dopant in the full supercell (e.g., $1/240 = 0.42$ at.% for the tetramer) is *not* the concentration used here; atomic percent is always dopants per lattice atom (n in $60n$). Tetramer α , gaps, and tetramer IPR/ J trends are not directly interchangeable with periodic $\mathcal{S}(n)$ without a matched-concentration benchmark; all $\mathcal{S}$ claims use the periodic protocol. In total, $\gtrsim 90$ tasks in Table III support the protocol-audited energy claims below; Marcus and Hirshfeld rows are separate diagnostics that do not enter Eq. (1). Charged-state tasks supply adiabatic IP/EA and vertical Marcus $\lambda^\pm$ (Supplemental Material; Sec. IIE 6), which do not enter Eq. (1).

D. Energy observables

Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain, tetramer *total* energy) is obtained from rigid PBE+D3 tetramer strain grids (Table VI alternate placement; reference-placement modernization in Table III). Periodic $\mathcal{S}$ and size-scaling energies use PBE+D3 throughout (Table III). Positive α means tension *raises* total energy; negative α means tension *lowers* it under the rigid-strain protocol. The raw substitution energy difference $\Delta E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at fixed strain is *not* a thermodynamic standard formation enthalpy (no elemental chemical potentials); when quoted, values use the same PBE+D3 protocol as the

surrounding grid. In sequential high-throughput screens one often estimates $E_{\text{tot}}(\epsilon, \delta) \approx E_{\text{sub}}(\epsilon=0) + E_{\text{strain}}(\epsilon)$; the neglected term is precisely $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. The order-of-magnitude ratio $|\mathcal{S}|/|E_{\text{sub}}|_{\text{per atom}}$ for screening workflows is discussed in Sec. IV as a rough magnitude estimate only (not a core nonlinearity test).

1. Non-additive coupling metric

We quantify strain–doping cross terms with the syn- order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times C_{60}$. The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization.

a. Relation to interaction and formation energies. Algebraically, Eq. (1) has the form of an interaction (coupling) energy, and we claim no novelty in that construction. Its distinct role is *dimensional*: a formation or substitution energy subtracts a single reference state, and a binding or interaction energy measures the coupling of one fixed pair of fragments, whereas $\mathcal{S}$ is a symmetric *double-reference cross term* over the four corners of the two-dimensional (ϵ, δ) control space that is spanned by high-throughput screening. It therefore quantifies a quantity none of the pairwise energies provide: the error incurred when total energies under joint load are assembled additively from separate strain-only and doping-only scans (Sec. III C). Table I contrasts $\mathcal{S}$ with common stability and electronic descriptors on these axes. Consistent with this purely energetic definition, $|\mathcal{S}|$ is *not* reducible to a single geometric ($\Delta \vec{d}$) or charge (Δq) channel (Sec. IV C 0 g); it is this irreducibility, rather than any new functional form, that motivates reporting $\mathcal{S}$ as a coupling descriptor for the joint design space.

When $\alpha_\delta = dE/d\epsilon$ denotes the *total-cell* linear strain coefficient (meV per 1% strain) at fixed n , a first-order estimate of the per-atom cross term is $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$ at small ϵ under the rigid-strain protocol; tetramer and periodic n use different N_{atoms} and dopant concentrations, so this link is not used quantitatively in the periodic audit (Discussion, argument scope).

Periodic-cell $\mathcal{S}$ values are extracted from CP2K total energies (Table III); the fifteen-point +3% strain audit accompanies the open data release. We report $\mathcal{S}$ for total energy only. By the same four-corner construction one may define $\mathcal{S}_{\text{gap}} = E_g(\epsilon, \delta) - [E_g(\epsilon, 0) + E_g(0, \delta) - E_g(0, 0)]$ (and analogously $\mathcal{S}_J$, $\mathcal{S}_\lambda$ for hopping or reorganization

energies); these observables are *not* tabulated here—panel (a) of Fig. 1 shows that gap slopes need not track $|\mathcal{S}|$.

b. Attribution framework (total energy). Conceptually, $\mathcal{S} = \mathcal{S}_{\text{geom}} + \mathcal{S}_{\text{el}} + \mathcal{S}_{\text{vdW}}$, where $\mathcal{S}_{\text{geom}}$ collects nonlinear elastic work from bond-angle and cage-shape changes beyond first-order strain, $\mathcal{S}_{\text{el}}$ from charge-transfer and orbital-rehybridization nonlinearities, and $\mathcal{S}_{\text{vdW}}$ from the Grimme D3 cross term. On the matched-functional tetramer P rigid four-corner grid, extracting D3 dispersion energies gives $\mathcal{S}_{\text{vdW}} \approx +0.33$ meV/atom—negligible versus $|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.2$ meV/atom (Table V). Corner-resolved $\mathcal{S}_{\text{vdW}}$ for B/N dopants and periodic cells beyond the matched tetramer P grid remain open; Table VII ($\vec{d}$, Δr_{cov}) and Hirshfeld population on the $n=1$ B/N/P strain paths (Sec. II E 4; Table III) proxy geometric vs. electronic channels in Discussion.

2. Localization and coupling (open data)

Following Capobianco *et al.* [4], inverse participation ratios and nearest-neighbor hopping J are archived for tetramer dimer fragments at the same PBE+D3 level as the energy scans; they are not used to infer $\mathcal{S}$.

3. Polaron binding

Marcus reorganization energies follow the vertical–adiabatic decomposition [4] (Supplemental Material). Charged geometry optimizations supply adiabatic IP/EA; vertical single-points at the neutral geometry yield Marcus $\lambda^\pm$.

E. Detailed calculation protocols

1. DFT calculator settings

Periodic total-energy and electronic-structure tasks use CP2K [18] with PBE [19] and Grimme DFT-D3 (Becke–Johnson damping) [20]. Fig. 1(c) uses rigid PBE+D3 α on the reference placement map (seed 42; 24/24; Table III). The alternate-placement benchmark (Table VI) tests slope sensitivity on an independent site map. The modernization grid mirrors the alternate-placement strain set ($\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$; pristine plus B/N/P rigid single-points on the archived reference map). Unless stated otherwise, biaxial strain scales the in-plane lattice vectors of the pristine equilibrium cell at fixed fractional coordinates (rigid-strain protocol). The plane-wave cutoff is 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table III); the atom-centered basis (DZVP-MOLOPT-SR-GTH) is unchanged across n . Tetramer models ($4 \times C_{60}$, 240 atoms) use B/N/P substitution at nominal 5% (12–13 dopant sites from archived

coordinates). Periodic supercells place one substituent per C_{60} unit ($n = 1-8$), giving n dopants among $60n$ lattice atoms (~ 1.67 at.% for all n).

2. Alternate-placement benchmark (Table VI)

The *reference* rows in Table VI reproduce an archived legacy-PBE rigid tetramer map (no D3) for placement contrast only; main-text quantitative claims use the alternate PBE+D3 column. The *alternate* placement repeats the full B/N/P strain grid ($\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$; 18 rigid single-point tasks) with an independently rerolled substitution map at the same nominal 5% doping level (pseudorandom seed 137; public archive). Reference-placement $\mathcal{S}_{+3\%}^{\text{tet}}$ for B/N/P is tabulated in Table VI and Sec. III C; periodic $\mathcal{S}(n)$ is not inferred from either tetramer map. Alternate tasks use PBE+D3 rigid single-point total-energy evaluations at 400 Ry with an SCF convergence threshold of $10^{-6} E_h$; they are *not* on the same exchange-correlation treatment as the reference α column, so only within-placement trends are compared once the grid converges. At periodic $n=4$, two independent site maps repeat the four-corner Eq. (1) audit (pristine and B/N/P at $\epsilon=0$ and $+3\%$; eight energies per map, 16 total) under the same PBE+D3 rigid protocol as the fifteen-point size-scaling grid; rankings of $|\mathcal{S}|$ at $+3\%$ are compared to the production reference map in Table II. Fixed pseudorandom seeds 137 and 271 identify the alternate maps in the archive; the production map uses seed 42. Where alternate-grid rigid single-points stall in SCF, the convergence threshold may be relaxed to $10^{-5} E_h$ for that task. The same relaxation applies to reference-placement modernization tasks when inner OT reaches 300 steps or outer SCF fails to converge (up to 21 outer cycles at $10^{-6} E_h$).

3. Ionic-relaxation benchmark (Table V)

The four tetramer corners in Table V are pristine and P -substituted cells at $\epsilon=0$ and $+3\%$ biaxial strain (the Eq. (1) grid for P). They use fixed-cell geometry optimization with BFGS (maximum 300 ionic steps for pristine corners and 450 for P -substituted corners) at prescribed in-plane cell vectors; only fractional coordinates relax while lattice vectors remain fixed. BFGS stops at 300 steps or when the maximum and RMS force components fall below 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr, or when the maximum and RMS atomic displacements fall below 3.0×10^{-3} and 1.5×10^{-3} bohr, respectively. These tasks use the same PBE+D3 settings as periodic audits: 400 Ry plane-wave cutoff, DZVP-MOLOPT-SR-GTH basis, and an SCF convergence threshold of $10^{-6} E_h$ (pristine) or $10^{-5} E_h$ (P -substituted) with OT-DIIS (up to 300 inner SCF iterations per ionic step and 20–40 outer SCF cycles) before updating the density for the next BFGS displacement. Rigid reference energies in

Table V remain the archived legacy-PBE (no D3) single-point grid; relaxed corners use PBE+D3, so the legacy cross-functional row is sign-qualitative only. A matched-functional control recomputes the same four corners as rigid PBE+D3 single points at the relaxed fractional coordinates (400 Ry, $10^{-6} E_h$ SCF, no ionic steps); quantitative retention is read from the matched-functional row in Table V, while the legacy-PBE rigid row remains for comparability with the archived tetramer grid.

4. Population strain paths ($n=1$ B/N/P)

Eighteen rigid single-point tasks on the $n=1$ B-, N-, and P -substituted periodic supercells scan Hirshfeld atomic charges along $\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$ (Table III); Mayer bond orders are extracted in post-processing from converged wavefunctions. These tasks use the same PBE+D3, 400 Ry, and $10^{-6} E_h$ SCF settings as the periodic size-scaling grid; doped $n=1$ cells have odd electron counts, so spin-polarized KS (LSD .TRUE.) is enabled (unlike the closed-shell size-scaling ENERGY tasks). They complement geometric $\bar{d}$ metrics in Table VII.

5. Periodic ionic-relaxation benchmark ($n=1$ P)

Fixed-cell geometry optimization on the periodic $n=1$ P -substitution four corners (pristine and P at $\epsilon=0$ and $+3\%$; PBE+D3, 400 Ry) uses the same BFGS thresholds as Table V. Inputs are generated from the converged rigid single-point cells in the size-scaling archive; all four corners converged ($\mathcal{S}_{\text{relaxed}} \ll \mathcal{S}_{\text{rigid}}$; Sec. III C 1)—do not rescale tetramer Table V benchmarks to periodic cells.

6. Marcus vertical step and post-processing

Charged-state geometry optimizations supply adiabatic ionization potentials and electron affinities ($E_{\text{opt}}^{\pm}$). Eight vertical single-point energy evaluations compute $E_{\text{vert}}^{\pm}$ at the *neutral*-geometry coordinates with the charged UKS settings of the relaxed state, following Ref. 4; $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$ (Supplemental Material). For near-gap B doping, negative λ^{-} from the vertical step at fixed neutral coordinates is treated as a semilocal artifact and excluded from rate estimates. π -projected densities of states use a 0.05 eV Gaussian broadening aligned to the Fermi level (Fig. 2). Inverse participation ratios and hopping integrals J are defined in Sec. II; verified IPR and J values accompany the open data release (not used to infer $\mathcal{S}$).

F. Validation benchmarks

The task-group inventory in Table III (superseded exploratory grids in Table IV) lists N_{conv} and main-text linkage; ionic-relaxation energies appear in Table V; the alternate-placement grid in Table VI spans the full B/N/P strain path and six modern PBE+D3 pristine references. Three validation benchmarks—ionic-relaxation tetramer $\mathcal{S}_P$ (Table V), alternate dopant placement (Table VI), and a cutoff benchmark at $6 \times \text{C}_{60} \text{ N @ } +3\%$ (Table III)—govern interpretation of size-scaling trends (Secs. III C and IV).

Sec. III B reports E_{sub} only when tied to the verified PBE+D3 grids above (no chemical-potential reference; not a formation enthalpy).

III. RESULTS

We organize the results around total-energy non-additivity (Sec. II B); Secs. III A–III C report band-edge context, PBE+D3 tetramer α and geometry (Tables VI and VII), and periodic $\mathcal{S}(n)$ (Tables III and II). Polaron transport diagnostics (Marcus λ , IPR, J) appear in the Supplemental Material and do not enter the $\mathcal{S}$ audit.

A. Electronic structure under strain and doping

Figure 1(a,b) summarizes tetramer HOMO–LUMO gaps (archived tetramer electronic-structure set) and representative π -DOS (B @ $\epsilon=0$; color scheme follows Ref. 4). At $\epsilon=0$, B is near a closed π gap ($E_g \approx 0.03$ eV), N doping pins the Fermi level inside the conduction band ($E_g \approx -0.14$ eV), and P remains gapped ($E_g \approx 0.05$ eV); tensile strain narrows the pristine gap (~ 23 meV/% on the three available strain points in the dataset). Donor vs. acceptor ordering of the $\epsilon=0$ gaps tracks distinct mechanical response in Sec. III B, but band-edge slopes are not used to infer $\mathcal{S}$. The four-panel π -DOS stack (Fig. 2) shows N-induced states crossing the Fermi level at $\epsilon=0$ (metallic pinning), a narrow π gap for B ($E_g \approx 0.03$ eV), and a wider gap for P—consistent with excess vs. deficient π electron count at the dopant cage but not yet decomposed into site-resolved Mulliken or Bader partitions (Limitations).

B. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/% on the verified PBE+D3 grids). Rigid PBE+D3 α on the reference placement map (seed 42; Table III) gives +58.7, −297.5, and +456.0 meV/% for B, N, and P; Fig. 1(c) plots this column, which shares the dopant-site map with the periodic $\mathcal{S}$ audit. On the independent alternate placement (Table VI), the same protocol gives +21.4, +45.4,

and +989.6 meV/%— α is placement-dominated while $\mathcal{S}_{+3\%}^{\text{tet}}$ remains $\mathcal{O}(10)$ meV/atom for all three dopants (Sec. IV D). The largest periodic $|\mathcal{S}|$ at $n=4$ and +3% strain is 23.7 meV/atom for P (Table III), while $|\alpha_P|$ on the alternate tetramer map is placement-dominated—foreshadowing α – $|\mathcal{S}|$ decoupling (Sec. III C). Figure 1(c) highlights the α spread; ranking dopants solely by substitution energy at $\epsilon=0$ is misleading once +3% strain is applied. Archived legacy-PBE single-point energies (no D3) are retained in the public release for reproducibility only and are not cited for quantitative claims in the main text. Table VII links mechanical spread to *tetramer* local geometry ($\sim 5\%$ nominal doping, multiple sites): P shows negligible strain-induced change in the mean nearest-neighbor C–C distance ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B at +3% load), indicating a vanishing *uniform nearest-neighbor stretch* component rather than absence of cage buckling; periodic $n=1$ bond statistics are not yet tabulated (Limitations). Complementary Hirshfeld population on the periodic $n=1$ B/N/P strain paths (Table III; Sec. II E 4) probes charge redistribution along the same load path as Table VII geometry.

C. Non-additive strain–doping coupling

Figure 1(d) tracks protocol $\mathcal{S}(n)$ at +3% biaxial strain. At $n = 4$ under the fixed periodic production map (Table II; two independent $n=4$ placement maps preserve the production-map $|\mathcal{S}|$ ranking at +3%: $P > B > N$), $\mathcal{S} = -6.2$ meV/atom (B), +3.0 meV/atom (N), and −23.7 meV/atom (P); P shows the largest magnitude despite α – $|\mathcal{S}|$ decoupling on the verified PBE+D3 tetramer map (Table VI). On the alternate tetramer map (Table VI, PBE+D3 throughout), P still carries the largest $\mathcal{S}_{+3\%}^{\text{tet}}$ (+41.4 meV/atom) while $|\alpha|$ is placement-dominated (+989.6 meV/%); this is *not* a simple α – $|\mathcal{S}|$ anticorrelation—decoupling means linear α cannot predict cross-term magnitude or sign (Sec. IV D). For $n \leq 4$ at matched 400 Ry cutoffs, $|\mathcal{S}|$ reaches 31.9 meV/atom for P at $n = 1$ and remains $\mathcal{O}(10)$ meV/atom for B and P at $n = 2$ –4 (Table II). Within $n \leq 4$, $|\mathcal{S}|$ is not monotonic in n (e.g., B: $8.4 \rightarrow 2.2 \rightarrow 6.2$ meV/atom at $n=1, 2, 4$), reflecting finite-cell dopant–dipole and image interactions rather than an asymptotic size limit; we report protocol values at each supercell without extrapolating $\mathcal{S}_\infty$. Rows with $n \geq 6$ use 350 Ry production settings and appear only in Supplemental Material Table S1. All periodic $\mathcal{S}$ values quoted in this subsection use the fixed dopant-site map and are tabulated in Table III (fifteen $n \times$ dopant points at +3% strain). Rigid PBE+D3 $\mathcal{S}_{+3\%}^{\text{tet}}$ from Eq. (1) on the alternate tetramer map is −2.6, +12.6, and +41.4 meV/atom for B, N, and P (Table VI); tetramer cross terms are not used to infer periodic $\mathcal{S}(n)$ because concentration and boundary conditions differ. Negative $\mathcal{S}$ (B, P at $n = 4$) means coupled tension *stabilizes* the doped cell relative to the additive estimate; positive $\mathcal{S}$ (N at $n = 4$) means sequen-

TABLE I. Positioning of the synergy order parameter $\mathcal{S}$ [Eq. (1)] against common stability and electronic descriptors. Algebraically $\mathcal{S}$ has the form of an interaction (coupling) energy; the distinction is *dimensional*, i.e. which axes of the joint (ϵ, δ) control space each quantity resolves. “Additive-screening error” denotes the bias incurred when total energies under joint load are assembled additively from separate strain-only and doping-only scans; $\mathcal{S}$ equals that bias by construction (Sec. II B). “Double-reference cross term” denotes a symmetric difference of differences over the four (ϵ, δ) corners rather than subtraction of a single reference state or fragment pair. Entries are conceptual scope, not numerical performance; consistent with its energetic definition, $|\mathcal{S}|$ is *not* reducible to a single geometric (Δd) or charge (Δq) channel (Sec. IV C 0 g).

Descriptor	Nonlinear (ϵ, δ) coupling	Additive-screening error	Double-reference cross term	Joint strain+doping
Formation / substitution energy E_{sub}	×	×	×	×
Binding / interaction energy ΔE_{int}	Partial	×	×	Partial
Linear strain coefficient α	×	×	×	Partial
Band gap E_g / work function	×	×	×	×
$\mathcal{S}$ (this work)	✓	✓	✓	✓

TABLE II. Periodic synergy audit at +3% biaxial strain (PBE+D3 rigid protocol, reference placement seed 42; core $n \leq 4$ at 400 Ry). $\mathcal{S}$ in meV/atom from Eq. (1). $n \geq 6$ rows (350 Ry production settings) are in Supplemental Material Table S1.

n (C ₆₀ units)	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
1	B	+3	−8.38
1	N	+3	+2.30
1	P	+3	−31.91
2	B	+3	−2.18
2	N	+3	+6.85
2	P	+3	−28.29
4	B	+3	−6.23
4	N	+3	+2.95
4	P	+3	−23.71

tial screens *overestimate* stability under load.

IV. DISCUSSION

A. Literature context

1. Ionic-relaxation checkpoint (Table V)

Table V benchmarks four fixed-cell geometry-optimization corners on the tetramer P -substitution grid (pristine and P at $\epsilon=0$ and +3%). For the pristine tetramer, relaxed total energies at $\epsilon=0$ and +3% are -1363.9551 and -1363.9550 E_h ($\Delta E \approx 1.2 \times 10^{-4}$ E_h), roughly an order of magnitude smaller than the archived rigid legacy-PBE nominal strain increment ($\sim 10^{-3}$ E_h). The P -doped corners give $E(P, \epsilon=0) = -1374.4415$ and $E(P, \epsilon=3\%) = -1374.4615$ E_h . Matched-functional PBE+D3 rigid single-points at relaxed coordinates yield $\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ meV/atom versus $\mathcal{S}_{\text{relaxed}} = -2.28$ meV/atom (sign *not* preserved; $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.9$; Table V). Matched-functional fixed-cell geometry optimization on the periodic $n=1$ P grid (PBE+D3; four corners) gives $\mathcal{S}_{\text{rigid}} = -31.9$ meV/atom versus $\mathcal{S}_{\text{relaxed}} \approx +0.1$ $\mu\text{eV/atom}$ ($|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}| \lesssim 10^{-5}$), so ionic relaxation collapses the rigid periodic cross term in this cell and rigid $|\mathcal{S}|$ is a conservative upper bound on equilibrium $\mathcal{S}$ at $n=1$. This tetramer benchmark does not rescale periodic $\mathcal{S}(n)$ (concentration and boundary conditions differ; Sec. IV).

Relative to strain-only, doping-only, mobility, or mechanical studies in qHP and related fullerene networks [1, 2, 4, 7–16, 21, 22], the distinction here is a periodic *total-energy* audit of joint biaxial (ϵ, δ) load [Eq. (1)] (periodic PBE+D3; Table III). Foundational DFT surveys of qHP C₆₀ monolayers emphasize anisotropic elastic moduli and optical gaps [8] rather than matched four-corner total-energy grids under joint substitutional and mechanical load. Strain-engineering roadmaps for generic two-dimensional materials [3] likewise foreground band-edge and mobility levers under separate mechanical or chemical protocols rather than a matched four-corner total-energy grid on covalent fullerene networks. Graphendofullerene derivatives report Curie temperatures tunable through *either* biaxial strain or electrostatic doping applied in isolation [12], reinforcing that joint (ϵ, δ) total-energy corners are not implicit in sequential protocol designs. In other 2D hosts, substitutional defects can raise or lower formation energies under biaxial strain depending on dopant–host size mismatch [22]; spin-resolved impurity engineering on qHP graphullerene instead targets half-semiconductor gaps [13]; we instead quantify the omitted additive cross term $\mathcal{S}$ in total energy. Band-edge metrics contextualize donor/acceptor

TABLE III. DFT task groups and main-text linkage (~ 90 tasks supporting quantitative claims). Superseded exploratory grids are listed in Table IV.

Task group	System	N_{conv}	Main text	Protocol
Archived baseline (legacy PBE)	$4 \times \text{C}_{60}$	24	Suppl. Mat.	Rigid strain; not in main-text claims
π -DOS and gaps	$4 \times \text{C}_{60}$	12	Fig. 2	PBE+D3
IPR (polaron dimer)	Dimer	4	Data release	Pristine/B; $\epsilon=0, +3\%$
Hopping J	Dimer	4	open data	PBE+D3; same corners
Charged polarons ^a	$4 \times \text{C}_{60}$	20	Suppl. Mat.	UKS; 12 GEO + 8 vert. SP
Periodic size scaling	$n \times \text{C}_{60}$	41	$\mathcal{S}(n)$	PBE+D3; 400/350 Ry
Cutoff benchmark	$6 \times \text{C}_{60}$ N	1	Sec. II F	+3%; 400 Ry control
Ionic relaxation	$4 \times \text{C}_{60}$	4	Table V	Fixed-cell GEO; P corners
Ionic relaxation	$1 \times \text{C}_{60}$ P	4	Sec. III C 1	Fixed-cell GEO; periodic
Alternate placement	$4 \times \text{C}_{60}$	24	Table VI	Rigid PBE+D3; alternate map
Periodic placement	$4 \times \text{C}_{60}$	16	Table II	$n=4$; two independent maps
Matched-functional rigid	$4 \times \text{C}_{60}$	4	Table V	PBE+D3 @ relaxed coords
Reference placement	$4 \times \text{C}_{60}$	24	Fig. 1(c)	Rigid PBE+D3; seed-42 map
Hirshfeld population	$1 \times \text{C}_{60}$ B/N/P	18	Discussion	Spin-polarized KS

^aSix relaxed tetramer geometries provide charged-state initial structures (not counted in N_{conv}).

TABLE IV. Superseded exploratory DFT grids (excluded from the ~ 90 -task count in Table III).

Task group	System	N_{conv}	Disposition
Early strain probe	$2 \times \text{C}_{60}$	5	Superseded
Single-molecule doping	C_{60}	11	Superseded
Early electronics	$2 \times \text{C}_{60}$	24	Superseded
Exploratory design grid	$4 \times \text{C}_{60}$	20	Archive only

ordering (Sec. III A) but are not used to infer $\mathcal{S}$.

B. Internal stress and external strain

Substitutional B, N, or P can be viewed as embedding a *localized internal strain field* ϵ_{int} into an already distorted qHP cage baseline [10]. Table VII quantifies the chemical-mismatch scale by covalent-radius excess Δr_{cov} (+7, −6, and +30 pm for B, N, and P vs. C) and by mean dopant–C distance $\bar{d}$ at $\epsilon=0$. External biaxial strain ϵ_{ext} is a uniform macroscopic load on the periodic cell. Sequential high-throughput models implicitly assume that responses to ϵ_{int} and ϵ_{ext} superpose linearly; $\mathcal{S}$ is the total-energy evidence that this superposition fails when both fields are present. In the Eshelby picture, a small substitutional defect in a linear elastic continuum produces a strain field that superposes with external load; qHP C_{60} violates that limit because the cage is *soft* and pre-strained [10], so second-order defect–elastic work (our geometric channel) can exceed linear $\alpha \times \epsilon$ scaling.

For *size-mismatched* P, the dopant introduces compressive local misfit in the cage ($\Delta r_{\text{cov}} = +30$ pm) while tensile ϵ_{ext} partially relieves that misfit without proportionally stretching nearest-neighbor bonds ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å at +3%; Table VII). The resulting negative $\mathcal{S}$ (cooperative stabilization relative to the additive estimate) is consistent with nonlinear *defect–elastic* coupling: external strain couples to *pre-existing cage buckling* rather than to uniform bond stretching. For B

and N, $|\Delta r_{\text{cov}}| \lesssim 10$ pm and $\Delta \bar{d}$ tracks ϵ_{ext} more uniformly ($\sim 8.5 \times 10^{-3}$ Å at +3%), so $\mathcal{S}$ is set primarily by *electronic* nonlinearity—donor vs. acceptor π counting and Fermi-level pinning (Secs. III A and III B)—rather than by large-amplitude cage reshaping. In this language, small-mismatch dopants approximate the linear-additive regime ($|\mathcal{S}| \sim \mathcal{O}(1\text{--}10)$ meV/atom at $n=1$), whereas large-mismatch substituents require explicit coupled (ϵ, δ) corners because geometric nonlinearity drives $|\mathcal{S}|$ to $\mathcal{O}(10^2)$ times the per-atom strain increment from α alone (Sec. III C).

a. Component attribution. Full decomposition into $\mathcal{S}_{\text{geom}}$, $\mathcal{S}_{\text{el}}$, and $\mathcal{S}_{\text{vdW}}$ awaits corner-resolved post-processing on every four-corner grid. For the matched-functional tetramer P rigid grid, extracting Grimme D3 dispersion energies at the four corners gives $\mathcal{S}_{\text{vdW}} \approx +0.33$ meV/atom—orders of magnitude below $|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.2$ meV/atom (Table V)—so vdW non-additivity does not drive the reported cross terms. Within the present audit, P combines negligible $\Delta \bar{d}$ with the largest $|\mathcal{S}|$ and a monotonic Hirshfeld decrease under tension on the $n=1$ strain path ($+0.063e \rightarrow +0.056e$ at +3%; Sec. IV C 0 g), implicating *pre-stored cage strain plus electronic redistribution* rather than uniform bond stretching as the dominant channel. B and N show smaller rigid $|\mathcal{S}|$ with modest Hirshfeld shifts over the same load path ($+0.12 \rightarrow +0.11e$ and $+0.069 \rightarrow +0.057e$ from $\epsilon=0$ to +3%; Table III), consistent with electronic-dominated cross terms at small Δr_{cov} .

TABLE V. Ionic-relaxation benchmark for the tetramer P -substitution synergy grid (four fixed-cell geometry-optimization corners: pristine and P at $\epsilon=0$ and $+3\%$, same biaxial cell). This benchmark is *sign-qualitative only*: the rigid reference uses legacy PBE without D3, whereas relaxed corners use PBE+D3, so the legacy cross-functional row is *not* a quantitative retention ratio. A matched-functional rigid reference (PBE+D3, same four corners): $\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ meV/atom; $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.9$ (sign *not* preserved on the matched-functional grid). We test whether $\mathcal{S}$ retains *sign within the tetramer grid only*; it does *not* calibrate periodic main-text $\mathcal{S}(n)$, which uses different dopant concentration and boundary conditions. Rigid reference energies: archived fixed-coordinate tetramer single-point grid (legacy PBE, no D3). Relaxed energies: PBE+D3 fixed-cell geometry optimization (400 Ry) from archived inputs in the open repository. $\mathcal{S}$ follows Eq. (1) with $N=240$ atoms.

Quantity	Rigid (legacy tetramer)	Relaxed (geometry opt.)
$E(\text{pristine}, \epsilon=0) (E_h)$	-1363.5512	-1363.9551
$E(\text{pristine}, \epsilon=3\%) (E_h)$	-1363.5522	-1363.9550
$E(P, \epsilon=0) (E_h)$	-1370.4965	-1374.4415
$E(P, \epsilon=3\%) (E_h)$	-1370.4890	-1374.4615
$\mathcal{S}$ (meV/atom)	+0.96	-2.28
Sign of $\mathcal{S}$ vs. rigid	+	- (not preserved)

TABLE VI. Tetramer rigid-strain placement sensitivity (PBE+D3 verified alternate map; archived legacy-PBE reference rows for placement contrast only). Main-text mechanical α and $\mathcal{S}_{+3\%}^{\text{tet}}$ use the *alternate* placement column (PBE+D3 throughout; Fig. 1(c)). The *reference* placement defines the periodic supercell site map (one substituent per C_{60}); its α and $\mathcal{S}_{+3\%}^{\text{tet}}$ rows reproduce an archived legacy-PBE grid (no D3) and are *not* used for quantitative claims in the main text. The alternate placement repeats the full B/N/P strain grid ($\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$; 18 rigid single-points) on an independently rerolled substitution map (Sec. II E 2). $\mathcal{S}_{+3\%}^{\text{tet}}$ follows Eq. (1) on the rigid tetramer four-corner grid ($N=240$). On the alternate map, large $|\alpha_P|$ and large $|\mathcal{S}_{+3\%}^{\text{tet}}|$ coexist—decoupling is *not* an α - $|\mathcal{S}|$ anticorrelation (Sec. IV D). Periodic $\mathcal{S}(n)$ is tabulated in Table III; the independent $n=4$ periodic placement audit and this alternate tetramer grid are discussed in Sec. II F.^a

Dopant	Placement	α (meV/%)	$\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom)	Status
B	reference	+56.9	+3.8	archived (legacy PBE)
N	reference	-303.2	+5.7	archived
P	reference	-19.1	+1.0	archived
B	alternate	+21.4	-2.6	verified (alternate)
N	alternate	+45.4	+12.6	verified (alternate)
P	alternate	+989.6	+41.4	verified (alternate)

^a Alternate site map: pseudorandom seed 137 (public archive).

C. Mechanistic synthesis

a. Opposite α for N vs. B. N introduces excess π electrons that strengthen bonding under tension (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α). Quantitatively, the $\epsilon=0$ gap ordering ($E_g^N < E_g^B < E_g^P$ within PBE; Sec. III A) and the dopant-resolved π -DOS stack in Figs. 1(b) and 2 (N states at E_F ; B near-gap; P gapped) align with donor vs. acceptor π counting, while total-energy α spans ~ 970 meV/% between B and P on the verified PBE+D3 alternate tetramer map (Table VI). Table VI repeats the full tetramer strain grid on an alternate site map (PBE+D3 rigid; Sec. II E 2): $|\alpha_P|$ exceeds B and N by $\mathcal{O}(10^2)$ while $\mathcal{S}_{+3\%}^{\text{tet}}$ at $+3\%$ remains $\mathcal{O}(10)$ meV/atom for all three dopants. Mayer bond order and Bader charge partitioning along the strain path are not yet computed (Limitations); periodic $\mathcal{S}(n)$ magnitudes on the reference map are not placement-averaged, though two independent $n=4$ placement maps preserve the $P > B > N$ ranking (Table III).

b. Large $|\mathcal{S}|$ for P despite small $|\alpha|$. P carries the largest covalent-radius excess (+30 pm vs. C; Table VII) and pre-existing cage distortion at $\epsilon=0$, yet shows the smallest strain-induced change in mean nearest C distance ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B/N) and the smallest site-to-site spread at $+3\%$ ($\sigma(\bar{d}) \approx 4 \times 10^{-4}$ Å vs. $\sim 2 \times 10^{-2}$ Å for B/N; Table VII). Tensile load therefore couples to *pre-existing* cage distortion from size mismatch rather than to a uniform elastic response, yielding a cross term invisible in separate scans but dominant when both strain and substitution are applied simultaneously—consistent with P carrying the largest $|\mathcal{S}|$ at $n=4$ (23.7 meV/atom; Table III) despite α - $|\mathcal{S}|$ decoupling on the verified PBE+D3 tetramer map (Table VI). Sec. III B presents this Table VII geometric decoupling before the $n=4$ $\mathcal{S}$ ranking in Sec. III C: tetramer $|\alpha|$ ranks P as weakly strain-sensitive, whereas the coupled periodic audit exposes the largest cross term.

c. N $\mathcal{S}$ sign change with n . With one substituent per C_{60} , electrostatic fields dilute at large n . The tetramer linear coefficient α and the periodic cross term $\mathcal{S}$ probe different energy partitions: α is the slope of *total-cell* energy under uniform lattice expansion at $\sim 5\%$

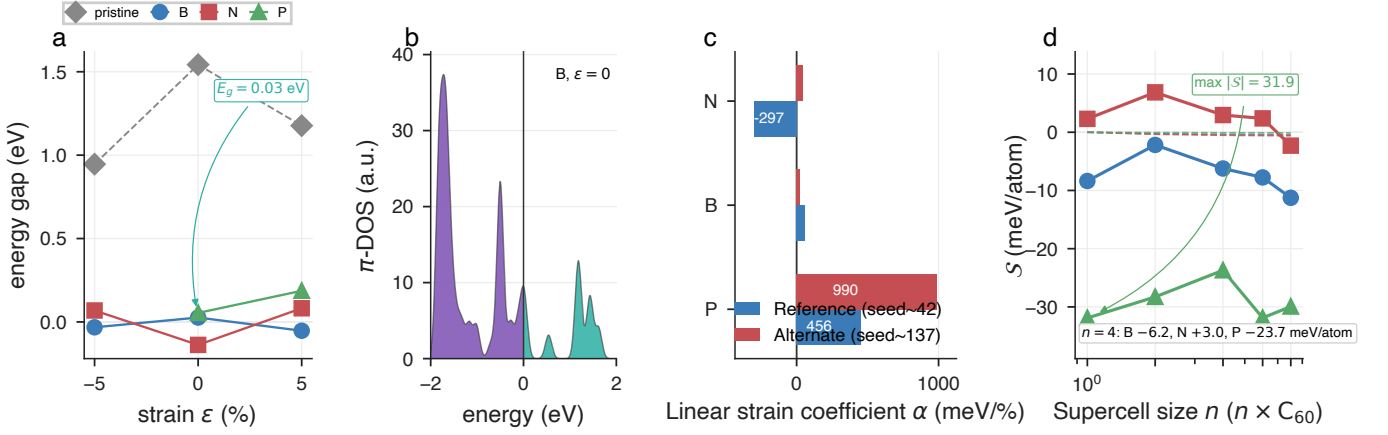


FIG. 1. Coupled (ϵ, δ) response in doped qHP C_{60} . (a) HOMO-LUMO gap vs. strain (tetramer electronic archive; PBE). (b) π -DOS for B @ $\epsilon=0$ (PBE+D3; N panel: $E_g < 0$ indicates metallic pinning). (c) Mechanical α on reference (blue) vs. alternate (red) tetramer placement maps (PBE+D3 rigid; Table III vs. Table VI). (d) Periodic $S(n)$ at +3% biaxial strain (PBE+D3 rigid; $n \leq 4$ core grid; peak $|S|$ at P, $n=1$).

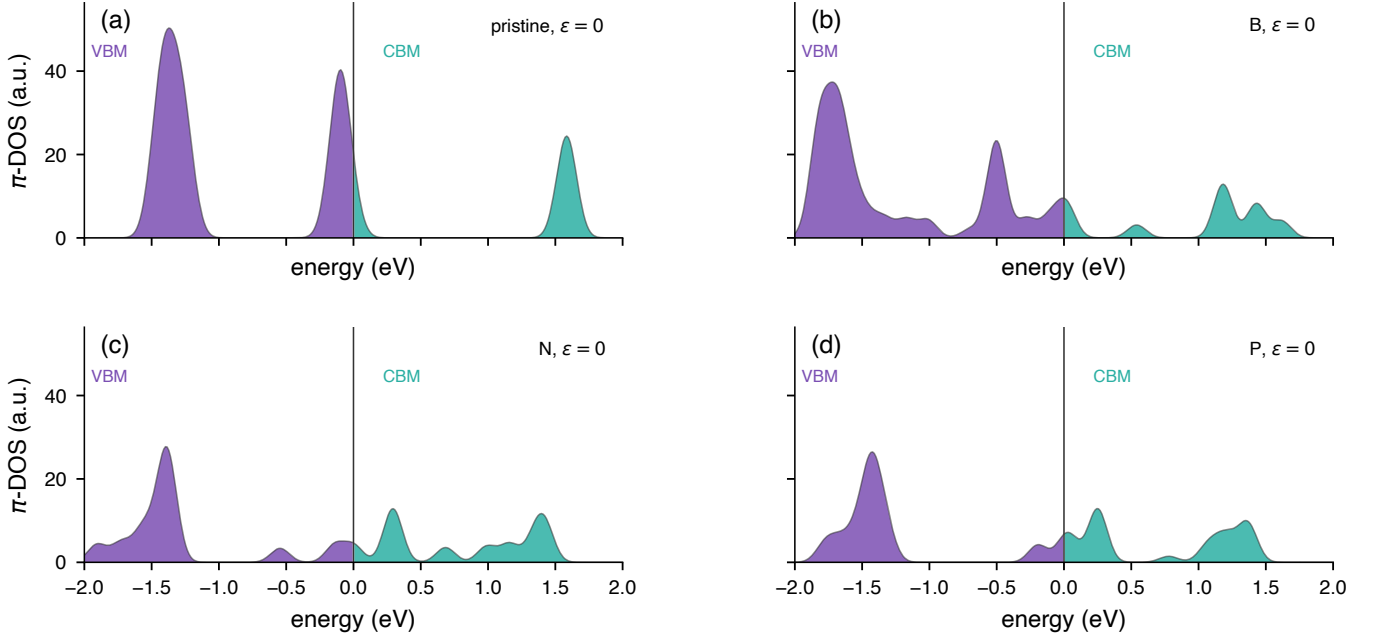


FIG. 2. π -projected density of states (π -DOS) for pristine and B/N/P-doped tetramers at $\epsilon=0$ (PBE+D3). Panels (a)–(d) share a common vertical scale; energy is relative to the Fermi level at 0 eV (Capobianco color scheme [4]).

multi-site doping, whereas S is the omitted four-corner cross term per atom at ~ 1.67 at.% under periodic boundary conditions. They therefore need not share sign even for the same element—a negative α (tension lowers total energy in a strain-only scan) can coexist with positive S (the coupled corner is *less* stable than an additive model predicts), as for N at $n=4$. Any N sign flip between $n=6$ and $n=8$ is confounded with the $400 \rightarrow 350$ Ry cutoff step and is excluded from design rules (Sec. IIF). Together, these three classes suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of α , *structural* local-distortion coupling to exter-

nal strain (P), and *finite-size* electrostatic dilution (N at large n).

d. *Geometric vs. electronic nonlinearity.* For P, Table VII shows that rigid tensile load barely moves the mean dopant-C distance ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å) yet S remains the largest at $n=4$ (23.7 meV/atom), whereas B/N exhibit $\sim 10^2 \times$ larger $\Delta \bar{d}$ at +3% but smaller $|S|$ (6.2 and 3.0 meV/atom). At $n=4$ the coupled correction for P exceeds the rigid per-atom strain increment from tetramer α by two orders of magnitude ($|S| \gg \alpha \times 3\% / N_{\text{atoms}}$; Methods energy observables), even though $|a|$ ranks P as the weakest strain responder on the tetramer grid.

TABLE VII. Local dopant–cage structure from archived *tetramer* XYZ ($\sim 5\%$ nominal doping; qualitative proxy for mismatch trends, not periodic $n=1$ bond statistics). Mean nearest-neighbor C distance $\bar{d}$ in Å; standard deviation $\sigma(\bar{d})$ across dopant sites; covalent-radius excess vs. C in pm. Strain-induced change $\Delta\bar{d} = \bar{d}(\epsilon=+3\%) - \bar{d}(\epsilon=0)$ tracks the *uniform nearest-neighbor stretch* component; cage buckling and angle distortions are not captured by $\bar{d}$ alone (Discussion Secs. IV B and IV C 0 g).

Dopant	$\bar{d}(\epsilon=0)$ (Å)	$\bar{d}(\epsilon=+3\%)$ (Å)	$\Delta\bar{d}$ (Å)	$\sigma(\bar{d})$ @ +3% (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	0.019	+7
N	1.393	1.402	+0.0087	0.024	−6
P	1.384	1.385	+0.00003	0.0004	+30

e. Argument scope (concentration and boundary conditions). We do *not* validate $\mathcal{S}$ by extrapolating tetramer α to periodic $n=4$: the $\sim 5\%$ tetramer vs. ~ 1.67 at.% periodic concentrations and tetramer vs. periodic boundary conditions are unmatched, so a first-order $(\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$ estimate is excluded from the argument. Non-additivity is assessed only through Eq. (1) on matched four-corner grids at common (ϵ, δ) ; P in particular requires explicit coupled evaluation rather than sequential α screening.

f. Sign origins of $\mathcal{S}$. For size-mismatched P (geometric channel), tensile ϵ_{ext} partially relieves dopant-induced cage compression; coupled corners are *more* stable than additive models predict, giving $\mathcal{S} < 0$ when the cross term is negative in our sign convention (B and P at $n=4$). For small-mismatch B and N (electronic channel), $\mathcal{S}$ follows nonlinear evolution of π states near E_F under strain: donor N and acceptor B occupy different filling regimes, so $\mathcal{S}$ can be positive or negative at comparable $|\Delta r_{\text{cov}}|$ (N +3.0 vs. B −6.2 meV/atom at $n=4$) without a single geometric scaling law.

g. Hirshfeld charge along the $n=1$ B/N/P strain paths. On the periodic $n=1$ P grid (PBE+D3), Hirshfeld charge on P decreases from +0.063 e at $\epsilon=0$ to +0.056 e at +3% (+0.050 e at +5%) while tetramer $\Delta\bar{d}$ for P remains below 10^{-4} Å at +3% load (Table VII). On the same periodic grid, B and N Hirshfeld charges shift from +0.119 e and +0.069 e at $\epsilon=0$ to +0.111 e and +0.057 e at +3% ($\sim 7\%$ and $\sim 18\%$ fractional $\Delta q/q$, vs. $\sim 11\%$ for P). Thus N shows the largest fractional charge response yet the smallest $|\mathcal{S}|$ at $n=4$, whereas P combines a moderate Δq with vanishing $\Delta\bar{d}$ and the largest cross term—charge redistribution alone does not rank $|\mathcal{S}|$; geometric pre-strain remains necessary for the P channel. Tensile strain therefore redistributes electron density without proportional nearest-neighbor bond stretching; Mayer bond orders along the same path are not yet tabulated (Limitations).

h. Significance versus DFT uncertainty. $\mathcal{S}$ is a difference of total energies computed with the *same* functional, basis, and cell for all four corners of Eq. (1), so systematic errors largely cancel in sign and relative magnitude. As a *rough magnitude estimate only* (not a core result), at $n=4$ and +3% strain P has $|\mathcal{S}| = 23.7$ meV/atom on the periodic PBE+D3 grid while tetramer and periodic concentrations differ ($\sim 5\%$ vs. ~ 1.67 at.%); $|\mathcal{S}|/|E_{\text{sub}}|_{\text{per atom}}$ ratios from archived

legacy grids are omitted because they mix functionals. Large absolute uncertainties in PBE E_{sub} therefore do not invalidate the conclusion that sequential scans omit a meV/atom-scale cross term.

i. Decomposition outlook. The stress-field picture (Sec. IV B) and Table VII–Hirshfeld trends indicate that $\mathcal{S}$ is not reducible to uniform bond stretching; Mayer/Bader corner projections remain open (Limitations).

D. Generality and design principles

a. Decoupling definition. α – $\mathcal{S}$ decoupling does *not* assert anticorrelation between $|\alpha|$ and $|\mathcal{S}|$: on the alternate PBE+D3 map (Table VI), P has both the largest $|\alpha|$ and the largest $\mathcal{S}_{+3\%}^{\text{tet}}$. The claim is that linear strain slopes from a one-parameter scan cannot quantitatively predict four-corner cross-term magnitude or sign because α is placement-sensitive at $\sim 5\%$ loading whereas $|\mathcal{S}|$ remains $\mathcal{O}(10)$ meV/atom on matched-functional tetramer and periodic audits.

b. Mismatch– $|\mathcal{S}|$ scaling (audit at $n=1, +3\%$). Using $|\Delta r_{\text{cov}}|$ from Table VII as a single mismatch coordinate, periodic $|\mathcal{S}|$ spans 2.3–8.4 meV/atom for B and N ($|\Delta r_{\text{cov}}| \lesssim 10$ pm) but reaches 31.9 meV/atom for P ($|\Delta r_{\text{cov}}|=30$ pm; Table II). N breaks a naive “ $|\mathcal{S}| \propto (\Delta r)^2$ ” law at comparable mismatch to B because its cross term is electronic (positive $\mathcal{S}$ at $n=4$) rather than geometric. The data nonetheless support a *two-regime* design rule for covalent networks: $|\Delta r_{\text{cov}}| \gtrsim 20$ pm flags the geometric-nonlinear regime where coupled audits are mandatory (P); $|\Delta r_{\text{cov}}| \lesssim 15$ pm leaves $|\mathcal{S}|$ at few-meV/atom under +3% load but still permits sign conflicts between α and $\mathcal{S}$ (B, N). When $|\mathcal{S}|$ approaches tens of meV/atom—as for P at $n=1$ –4—stability margins inferred from $\epsilon=0$ substitution energies or strain-only α can shift by several percent of the per-atom substitution scale even without reordering dopants (Sec. IV C 0 h).

c. When additive scans suffice. **Dopant mismatch:** at $|\Delta r_{\text{cov}}| \lesssim 10$ pm and +3% biaxial load, $|\mathcal{S}| \lesssim 10$ meV/atom on the $n=1$ audit grid; sequential pipelines may be adequate for coarse ranking but not for close E_{sub} competitors. **Strain magnitude:** pristine tetramers show $|\alpha| \approx 1$ meV/%, so sub-1% loads keep linear strain increments small; at +3% and above, coupled corners for size-mismatched dopants are advis-

able because $|\mathcal{S}|$ can exceed the rigid per-atom increment from $\alpha \times \epsilon$ by orders of magnitude (P at $n=4$). **Network architecture:** qHP C_{60} combines interfullerene covalent links with *soft* intramolecular cages and intrinsic intramolecular strain [10], amplifying local-distortion coupling relative to rigid 2D lattices; extrapolation to graphene-like hosts requires separate benchmarks (Limitations).

d. Co-design implications. At $n=4$ and +3% strain the coupled correction is $|\mathcal{S}| = 23.7$ meV/atom for P vs. 6.2 meV/atom for B—sequential *additive* estimates differ by approximately a factor of four in the load-dependent *margin* under the same PBE+D3 periodic protocol. Dopants with covalent-radius excess $\gtrsim 20$ pm relative to C (P: +30 pm vs. B: +7 pm; Table VII) show the largest protocol $|\mathcal{S}|$ under fixed coordinates; *size-mismatched* substituents should be screened with coupled (ϵ, δ) total-energy evaluations rather than sequential one-parameter pipelines. The fifteen-point periodic grid tabulates protocol $|\mathcal{S}|$ at each (n, δ) as an audit of additive omission (Table III), not as quantitative design targets. At +3% strain, P carries the largest additive-approximation error in protocol $\mathcal{S}$ ($|\mathcal{S}| = 23.7$ – 31.9 meV/atom from $n=4$ to $n=1$ under fixed coordinates); qualitative co-design should treat P under tension as a *coupled* stability problem (Table VI shows $|\alpha_P|$ placement-dominated on the alternate map), not $|E_{\text{sub}}|$ at $\epsilon = 0$ alone. Where $\mathcal{S} < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $\mathcal{S} > 0$ (N at $n=4$), sequential screens overestimate stability under load. For B at $n=4$, negative $\mathcal{S}$ coexists with positive α —a sign conflict that strain-only elastic estimates cannot resolve without evaluating $\mathcal{S}$. For N at $n=4$, negative tetramer α and positive $\mathcal{S}$ present a second sign conflict at ~ 1.67 at.% periodic doping. Gap closing [Fig. 1(a)] does not track $|\mathcal{S}|$; $\mathcal{S}_{\text{gap}}$ is defined in Sec. II but not computed, so photonic or transport targets cannot be inferred from total-energy $\mathcal{S}$ alone.

E. Experimental signatures

Three classes of laboratory observables could test the stress-field picture without resolving total energies directly. **Raman or IR cage modes:** under joint doping and strain, mode shifts should deviate from the sum of doping-only and strain-only shifts when $|\mathcal{S}|$ is large—especially for P, where tensile load couples to pre-existing cage distortion rather than uniform bond stretching (Table VII). **Scanning tunneling spectroscopy:** dopant-site local density of states should evolve nonlinearly with strain when Hirshfeld charge redistributes without proportional $\Delta\bar{d}$ (Sec. IV C 0g); N pinning near E_F vs. gapped B/P provides contrasting signatures. **Thermal or chemical stability:** for $\mathcal{S} < 0$ (B, P at $n=4$), tension should enhance apparent doping stability relative to additive thermochemical estimates; for $\mathcal{S} > 0$ (N at $n=4$), the opposite bias is expected. Quantitative links

require matched sample chemistry and strain calibration; we quote DFT protocol bounds only.

F. Validation protocol (Tables III–VI)

Sec. III C 1 reports the tetramer ionic-relaxation benchmark (Table V) and periodic $n=1$ P fixed-cell geometry optimization ($\mathcal{S}_{\text{relaxed}} \ll \mathcal{S}_{\text{rigid}}$). The alternate placement map (Table VI, PBE+D3 rigid grid) gives $\alpha = +21.4, +45.4$, and $+989.6$ meV/% and $\mathcal{S}_{+3\%}^{\text{tet}} = -2.6, +12.6$, and $+41.4$ meV/atom for B, N, and P. Two independent $n=4$ periodic placement maps preserve the production-map $|\mathcal{S}|$ ranking at +3% (P>B>N; Table III). The cutoff benchmark row in Table III tests 400 Ry vs. 350 Ry for $6 \times \text{C}_{60}$ N @ +3%.

G. Limitations and outlook

Rigid-strain periodic $\mathcal{S}(n)$ and tetramer α both use PBE+D3 in the main-text quantitative audit (Tables III and VI); archived legacy-PBE grids in the public release are not used for those claims. Fixed-cell ionic relaxation is reported for the tetramer P grid (Table V) and for periodic $n=1$ P (four PBE+D3 corners; Sec. III C 1), where $\mathcal{S}_{\text{relaxed}} \ll \mathcal{S}_{\text{rigid}}$. All periodic $\mathcal{S}(n)$ use a single production dopant-site map; magnitudes are configuration-specific (two independent $n=4$ placement maps preserve the P>B>N ranking; Table III). Production $n \geq 6$ points use 350 Ry while core claims require matched 400 Ry (cutoff benchmark in Table III). Additional limits: tetramer vs. periodic concentration mismatch; Table V tetramer geometry is a qualitative proxy; PBE+D3 without Koopmans/rVV10; $\mathcal{S}_{\text{gap}}$ not quantified; Mayer bond orders not tabulated; Hirshfeld population covers $n=1$ B/N/P (Sec. II E 4); unphysical B λ^- under the fixed neutral-geometry vertical protocol (Supplemental Material) is excluded from rate estimates; mobility tensors are not computed. Future work adds Mayer/Bader strain-path partitioning and placement averaging on periodic cells.

V. CONCLUSIONS

We quantify how substitutional chemistry and biaxial strain couple *nonlinearly* in covalent molecular networks: the omitted total-energy cross term $\mathcal{S}$ arises when a dopant-induced internal stress field meets an external load, not from a graphullerene-specific numerical artifact. On qHP C_{60} as the validation platform, two mechanisms emerge—*electronic* donor/acceptor π control (B, N) and *geometric* local-distortion coupling for size-mismatched P—with a practical mismatch threshold ($|\Delta r_{\text{cov}}| \gtrsim 20$ pm) separating few-meV from tens-of-meV $|\mathcal{S}|$ under +3% load (Secs. IV B and IV D). In detail:

1. distinct donor/acceptor π filling and gap response (Sec. III A) with placement-sensitive PBE+D3 α on tetramer maps (Table VI);
2. meV/atom-scale omitted cross terms on a fifteen-point periodic grid ($|\mathcal{S}|$ up to 31.9 meV/atom for P at $n=1$ at fixed coordinates; Table III);
3. $|\mathcal{S}|$ tracking local cage distortion and mismatch (Table VII) rather than tetramer $|\alpha|$ for P, including on the PBE+D3 alternate tetramer map (Table VI).

Under the rigid-strain protocol (Sec. II B), quoted $|\mathcal{S}|$ magnitudes bound equilibrium cross terms; periodic $n=1$ P ionic relaxation nearly extinguishes $\mathcal{S}$ ($|\mathcal{S}_{\text{rel}}| \ll 1$ meV/atom vs. 31.9 meV/atom rigid), while the tetramer Table V benchmark shows sign reversal at matched corners. Coupled (ϵ, δ) screening is advisable for size-mismatched substituents, +3% and larger loads, and close substitution-energy competitors; Raman, STS, and stability trends in Sec. IV E offer experimental tests of the stress-field picture.

ACKNOWLEDGMENTS

We acknowledge computational resources and support from Xiamen University.

DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/chenxingqiang/graphullerene-strain-engineering>, including computational inputs and outputs, tabulated synergy and polaron analyses, and materials to reproduce all figures and tables in the main text. Ionic-relaxation, alternate tetramer placement, periodic placement, Hirshfeld population, and size-scaling archives referenced in Tables III–VI are included in the public release. Fixed pseudorandom seeds identifying alternate site maps (e.g. seed 137 for Table VI) are recorded in the archive metadata. A tagged release matching the manuscript revision will be issued upon acceptance.

-
- [1] S. Chen, H. Liu, and K. Wang, untangling the respective effects of heteroatom doping and strain on electronic transport in 2D materials, *Advanced Materials* **36**, 2301234 (2024).
 - [2] H. Li, M. A. Shakoori, and R. Wang, strain engineering of electronic structure and thermoelectric properties of quasi-hexagonal fullerene monolayer, *J. Appl. Phys.* **136**, 015101 (2024).
 - [3] A. K. Katiyar and J. H. Ahn, strain-engineered 2D materials: challenges, opportunities, and future perspectives, *Small Methods* **9**, 2401404 (2025).
 - [4] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa, electron localization and mobility in monolayer fullerene networks, *Nano Lett.* **24**, 7891 (2024).
 - [5] Y. Yang, W. Zhang, H. Liu, and K. Li, two-dimensional carbon allotropes with tunable direct band gaps and high carrier mobility, *Appl. Surf. Sci.* **536**, 147815 (2021).
 - [6] P. Silva, R. M. Faria, L. G. Ferreira, and J. L. F. Fatima, large polarons in two-dimensional fullerene networks: the crucial role of anisotropy in charge transport, *Nanoscale* **16**, 2337 (2024).
 - [7] Y.-B. Shi, S.-H. Lv, Z.-F. Shao, H.-K. Dong, S. Cao, and P. Qian, a first-principles study of 1d and 2d C₆₀ nanostructures: strain effects on band alignments and carrier mobility, *J. Phys.: Condens. Matter* **35**, 225701 (2023).
 - [8] R. M. Tromer, L. A. R. Junior, and D. S. Galvao, A DFT study of the electronic, optical, and mechanical properties of a recently synthesized monolayer fullerene network, *Chemical Physics Letters* **804**, 139925 (2022).
 - [9] R. Lv, J. Xi, M. Chen, and S. Yang, Covalent fullerene networks: preparation, properties and catalytic applications, *Rev. Mater. Res.* **2**, 100195 (2026).
 - [10] G. Makov and U. Argaman, structure and properties of graphullerene: a semiconducting two-dimensional C₆₀ crystal, *npj Comput. Mater.* **9**, 211 (2023).
 - [11] B. Peng and M. Pizzochero, monolayer C₆₀ networks: a first-principles perspective, *Chem. Commun.* **61**, 10287 (2025).
 - [12] D. López-Alcalá, Z. Hu, and J. J. Baldoví, graphendo-fullerene: a novel molecular two-dimensional ferromagnet, *Chemical Science* **16**, 7659 (2025).
 - [13] M. A. Khan and M. N. Leuenberger, tuning the electronic and optical properties of impurity-engineered two-dimensional graphullerene half-semiconductors, *Phys. Rev. Mater.* **9**, 034001 (2025).
 - [14] J. Wang, K. Li, and H. Zhang, simulation on stress-related anisotropy of qTP C₆₀ and qHP C₆₀: implications for optoelectronic nanodevices, *ACS Applied Nano Materials* **7**, 3456 (2024).
 - [15] F. Qiu, H. Wang, P. Zhang, and Y. Li, atomic-scale insights into the mechanical and tribological properties of graphullerene, *Tribology International* **204**, 110756 (2025).
 - [16] M. L. P. Junior, R. M. Tromer, L. A. R. Junior, and D. S. Galvão, endohedral derivatives of the recently synthesized two-dimensional fullerene networks: electronic and optical insights from first-principles calculations, *arXiv preprint* (2026), arXiv:2603.10142 [cond-mat.mtrl-sci].
 - [17] M. Alihosseini and M. Neek-Amal, Strain tuning of optoelectronic properties of covalent organic framework bilayers and heterostructures, *Phys. Rev. B* **108**, 165403 (2023).
 - [18] J. VandeVondele, M. Krack, F. Mohamed, M. Parrinello, T. Chassaing, and J. Hutter, Quickstep: Fast and accurate density functional calculations using a mixed gaussian and plane waves approach, *Computer Physics Communications* **167**, 103 (2005).

- [19] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized gradient approximation made simple, *Phys. Rev. Lett.* **77**, 3865 (1996).
- [20] S. Grimme, S. Ehrlich, and L. Goerigk, effect of the damping function in dispersion corrected density functional theory, *Journal of Computational Chemistry* **32**, 1456 (2011).
- [21] A. Shaikh, J. Wu, and B. Peng, negative and positive anisotropic thermal expansion in 2D fullerene networks, *arXiv preprint* (2025), arXiv:2504.02037 [cond-mat.mtrl-sci].
- [22] P. Santra, S. Ghaderzadeh, M. Ghorbani-Asl, H.-P. Komsa, E. Besley, and A. V. Krashennnikov, strain-modulated defect engineering of two-dimensional materials, *npj 2D Materials and Applications* **8**, 33 (2024).